Erratum to the explicit transfer matrices of M, and tight asymptotics of $R(E^N)$, $\mathrm{Tr}(M)$ and the k-cycle variance

June 12, 2026

Conventions. Replica patterns of an index quadruple (a, b, c, d): $4K$: all equal; A: $a{=}b \neq c{=}d$; U_1: $a{=}c \neq b{=}d$; U_2: $a{=}d \neq b{=}c$. All vectors and matrices are indexed in the order $(A, U_1, U_2, 4K)$. Bulk matrices are written $W^P[Z_{\mathrm{in}}, Z_{\mathrm{out}}]$ (Z_{in} a fixed representative on the incoming bond, Z_{out} summed over its exact pattern class), and chain values are $u_{\mathrm{init}}^{\top} W^{P_3} \cdots W^{P_{N-1}} u_{\mathrm{end}}$; this orientation is fixed by the validation below. $\sigma_2 = \mathbb{E}[Q^2]$, $\sigma_4 = \mathbb{E}[Q^4]$ for a core entry Q, $c := \sigma_4/\sigma_2^2$ ($c{=}3$ Gaussian, $c{=}1$ e.g. Rademacher), $m = m_1 m_2$, and the unbiasedness normalization is $\sigma_2 = 1/m$.

1 Erratum: explicit entries of W^{4K}, W^{U_1}, W^A and three initial vectors

The explicit polynomial entries in the document disagree with the construction they describe in **33 places** (under $m = m_1 m_2$). This was established three independent ways, all agreeing to machine precision: (i) dense Monte–Carlo simulation of the actual sketch ($\geq 5\sigma$ discrepancies at $N \geq 4$); (ii) an exact 4-replica Gaussian–Wick evaluator (no pattern formalism); (iii) an exact symbolic partition-sum engine over the 15^3–15^4 set-partition triples/quadruples per site, which reproduces (ii) exactly and (i) within Monte-Carlo error. The *theorem-level* statements (per-entry chain decomposition, aggregation identities, $\|M\|_\infty = R(E^N)$, the eigenvector lemma) are not contradicted; only the explicit polynomials, and hence every numerical consequence, change. All u_{end} vectors and the initial vectors of types $(A, \cdot), (\cdot, A), (4K, 4K)$ are *exact* as published; consequently the $N{=}3$ value of $\mathrm{Tr}(M)$ was correct while everything at $N \geq 4$, and $R(E^N)$ already at $N{=}3$, was not.

object	wrong entries $(Z_{\mathrm{in}}, Z_{\mathrm{out}})$	leading relative si
W^{4K}	$(A,4K)$, $(U_1,4K)$, $(U_2,4K)$, $(4K,A)$, $(4K,U_1)$, $(4K,U_2)$, $(4K,4K)$	$\Theta(1)$ on the $4K$ c
W^{U_1}	(A,U_1), $(A,4K)$, (U_1,A), (U_1,U_2), $(U_1,4K)$, (U_2,U_1), $(U_2,4K)$, $(4K,A)$, $(4K,U_1)$, $(4K,U_2)$	$\Theta(1)$ on the $4K$ c
W^{A}	(A,U_1), (A,U_2), $(A,4K)$, (U_1,A), $(U_1,4K)$, (U_2,A), $(U_2,4K)$, $(4K,A)$, $(4K,U_1)$, $(4K,U_2)$	$\Theta(1)$ on the $4K$ c
$u_{\mathrm{init}}^{(4K,U_1)}$	$[U_1]$	$\Theta(1/m)$ relative
$u_{\mathrm{init}}^{(U_1,4K)}$	$[4K]$	$\Theta(1/m)$ relative
$u_{\mathrm{init}}^{(U_1,U_1)}$	$[A]$, $[U_1]$, $[U_2]$, $[4K]$	$\Theta(1/m)$ relative

Table 1: Erratum census. The $\{A, U_1, U_2\}^2$ block of W^{4K}, six entries of each of W^{U_1}, W^A, all u_{end}, and all A-type/$(4K, 4K)$ initial vectors are correct as published. The $(m_1{-}m_2)$-proportional errors vanish for balanced splits; the $4K$-channel errors do not.

1.1 Corrected entries (machine-generated from the validated engine; expressed in (m_1, m_2) with $m = m_1 m_2$)

W^{4K}.

$$\begin{aligned}\left[W^{4K}\right]_{A,4K} = m_1^3 m_2^3 \sigma_2^2 \Big[& 1\,\sigma_4^2 \\ & + \sigma_2^2 \left(m_1 m_2 + m_1 - 2\right)\,\sigma_4 \\ & + \sigma_2^4 \left(m_1^2 m_2^2 - m_1 m_2 - m_1 + 2m_2 - 1\right) \Big]\end{aligned}$$

$$\begin{aligned}\left[W^{4K}\right]_{U_1,4K} = m_1^3 m_2^3 \sigma_2^2 \Big[& 1\,\sigma_4^2 \\ & + \sigma_2^2 \left(m_1 m_2 + m_1 - 2\right)\,\sigma_4 \\ & + \sigma_2^4 \left(m_1^2 m_2^2 - m_1 m_2 - m_1 + 2m_2 - 1\right) \Big]\end{aligned}$$

$$\begin{aligned}\left[W^{4K}\right]_{U_2,4K} = m_1^3 m_2^3 \sigma_2^2 \Big[& 1\,\sigma_4^2 \\ & + \sigma_2^2 \left(m_1 m_2 + m_1 - 2\right)\,\sigma_4 \\ & + \sigma_2^4 \left(m_1^2 m_2^2 - m_1 m_2 - m_1 + 2m_2 - 1\right) \Big]\end{aligned}$$

$$\begin{aligned}\left[W^{4K}\right]_{4K,A} = m_1^3 m_2^3 \sigma_2^2 \Big[& 1\,\sigma_4^2 \\ & + \sigma_2^2 \left(m_1 m_2 + m_2 - 2\right)\,\sigma_4 \\ & + \sigma_2^4 \left(m_1^2 m_2^2 - m_1 m_2 + 2m_1 - m_2 - 1\right) \Big]\end{aligned}$$

$$\begin{aligned}\left[W^{4K}\right]_{4K,U_1} = m_1^3 m_2^3 \sigma_2^2 \Big[& 1\,\sigma_4^2 \\ & + \sigma_2^2 \left(m_1 m_2 + m_2 - 2\right)\,\sigma_4 \\ & + \sigma_2^4 \left(m_1^2 m_2^2 - m_1 m_2 + 2m_1 - m_2 - 1\right) \Big]\end{aligned}$$

$$\begin{aligned}\left[W^{4K}\right]_{4K,U_2} = m_1^3 m_2^3 \sigma_2^2 \Big[& 1\,\sigma_4^2 \\ & + \sigma_2^2 \left(m_1 m_2 + m_2 - 2\right)\,\sigma_4 \\ & + \sigma_2^4 \left(m_1^2 m_2^2 - m_1 m_2 + 2m_1 - m_2 - 1\right) \Big]\end{aligned}$$

$$\begin{aligned}\left[W^{4K}\right]_{4K,4K} = m_1^3 m_2^3 \Big[& 1\,\sigma_4^3 \\ & + \left(3\sigma_2^4 \left(m_1 m_2 + m_1 + m_2 - 3\right)\right)\sigma_4 \\ & + \left(3\sigma_2^6 \left(m_1^2 m_2^2 - m_1 m_2 - m_1 - m_2 + 2\right)\right) \Big]\end{aligned}$$

W^{U_1}.

$$\left[W^{U_1}\right]_{A,U_1} = m_1^3 m_2^4 \sigma_2^6 \left(m_1 m_2 - 1\right)$$

$$\left[W^{U_1}\right]_{A,4K} = m_1^3 m_2^3 \sigma_2^4 \Big[1 \, \sigma_4 + \sigma_2^2 \left(m_1 m_2 + m_2 - 2\right) \Big]$$

$$\left[W^{U_1}\right]_{U_1,A} = m_1^4 m_2^3 \sigma_2^6 \left(m_1 m_2 - 1\right)$$

$$\left[W^{U_1}\right]_{U_1,U_2} = m_1^4 m_2^3 \sigma_2^6 \left(m_1 m_2 - 1\right)$$

$$\left[W^{U_1}\right]_{U_1,4K} = m_1^4 m_2^3 \sigma_2^4 \Big[1 \, \sigma_4 + \sigma_2^2 \left(m_1 m_2^2 - 1\right) \Big]$$

$$\left[W^{U_1}\right]_{U_2,U_1} = m_1^3 m_2^4 \sigma_2^6 \left(m_1 m_2 - 1\right)$$

$$\left[W^{U_1}\right]_{U_2,4K} = m_1^3 m_2^3 \sigma_2^4 \Big[1 \, \sigma_4 + \sigma_2^2 \left(m_1 m_2 + m_2 - 2\right) \Big]$$

$$\left[W^{U_1}\right]_{4K,A} = m_1^3 m_2^3 \sigma_2^4 \Big[1 \, \sigma_4 + \sigma_2^2 \left(m_1 m_2 + m_1 - 2\right) \Big]$$

$$\left[W^{U_1}\right]_{4K,U_1} = m_1^3 m_2^4 \sigma_2^4 \Big[1 \, \sigma_4 + \sigma_2^2 \left(m_1^2 m_2 - 1\right) \Big]$$

$$\left[W^{U_1}\right]_{4K,U_2} = m_1^3 m_2^3 \sigma_2^4 \Big[1 \, \sigma_4 + \sigma_2^2 \left(m_1 m_2 + m_1 - 2\right) \Big]$$

W^{A}.

$$\left[W^{A}\right]_{A,U_1} = m_1^4 m_2^3 \sigma_2^6 \left(m_1 m_2 - 1\right)$$

$$\left[W^{A}\right]_{A,U_2} = m_1^4 m_2^3 \sigma_2^6 \left(m_1 m_2 - 1\right)$$

$$\left[W^A\right]_{A,4K} = m_1^4 m_2^3 \sigma_2^4 \Big[1 \; \sigma_4 + \sigma_2^2 \left(m_1 m_2^2 - 1 \right) \Big]$$

$$\left[W^A\right]_{U_1,A} = m_1^3 m_2^4 \sigma_2^6 \left(m_1 m_2 - 1 \right)$$

$$\left[W^A\right]_{U_1,4K} = m_1^3 m_2^3 \sigma_2^4 \Big[1 \; \sigma_4 + \sigma_2^2 \left(m_1 m_2 + m_2 - 2 \right) \Big]$$

$$\left[W^A\right]_{U_2,A} = m_1^3 m_2^4 \sigma_2^6 \left(m_1 m_2 - 1 \right)$$

$$\left[W^A\right]_{U_2,4K} = m_1^3 m_2^3 \sigma_2^4 \Big[1 \; \sigma_4 + \sigma_2^2 \left(m_1 m_2 + m_2 - 2 \right) \Big]$$

$$\left[W^A\right]_{4K,A} = m_1^3 m_2^4 \sigma_2^4 \Big[1 \; \sigma_4 + \sigma_2^2 \left(m_1^2 m_2 - 1 \right) \Big]$$

$$\left[W^A\right]_{4K,U_1} = m_1^3 m_2^3 \sigma_2^4 \Big[1 \; \sigma_4 + \sigma_2^2 \left(m_1 m_2 + m_1 - 2 \right) \Big]$$

$$\left[W^A\right]_{4K,U_2} = m_1^3 m_2^3 \sigma_2^4 \Big[1 \; \sigma_4 + \sigma_2^2 \left(m_1 m_2 + m_1 - 2 \right) \Big]$$

$u_{\text{init}}^{(4K,U_1)}$.

$$\left[u_{\text{init}}^{(4K,U_1)}\right]_{U_1} = m_1^4 m_2^5 \sigma_2^4 \Big[1 \; \sigma_4^2 + \sigma_2^2 \left(m_1^2 m_2 - 1 \right) \sigma_4 + \left(\sigma_2^4 \left(m_1 m_2 - 1 \right) \left(m_1^2 m_2 + 2 \right) \right) \Big]$$

$u_{\text{init}}^{(U_1,4K)}$.

$$\left[u_{\text{init}}^{(U_1,4K)}\right]_{4K} = m_1^4 m_2^4 \sigma_2^2 \Big[1 \; \sigma_4^3 + \sigma_2^2 \left(m_1 m_2 - 1 \right) \sigma_4^2 + \sigma_2^4 \left(m_1^2 m_2^2 + m_1^2 m_2 + 2 m_1 + 3 m_2 - 7 \right) \sigma_4 + \sigma_2^6 \left(m_1^3 m_2^3 + m_1^2 m_2^2 - m_1^2 m_2 + 2 m_1 m_2^2 - 3 m_1 m_2 - 2 m_1 - 5 m_2 + 7 \right) \Big]$$

$u_{\text{init}}^{(U_1,U_1)}$.

$$\left[u_{\text{init}}^{(U_1,U_1)}\right]_A = m_1^4 m_2^4 \sigma_2^6 \Big[\, 1 \;\; \sigma_4 \;\; + \sigma_2^2 \left(m_1^2 m_2 + m_1 m_2 - 2\right) \Big]$$

$$\left[u_{\text{init}}^{(U_1,U_1)}\right]_{U_1} = m_1^4 m_2^5 \sigma_2^6 \Big[\, 1 \;\; \sigma_4 \;\; + \sigma_2^2 \left(m_1^3 m_2^2 - 1\right) \Big]$$

$$\left[u_{\text{init}}^{(U_1,U_1)}\right]_{U_2} = m_1^4 m_2^4 \sigma_2^6 \Big[\, 1 \;\; \sigma_4 \;\; + \sigma_2^2 \left(m_1^2 m_2 + m_1 m_2 - 2\right) \Big]$$

$$\left[u_{\text{init}}^{(U_1,U_1)}\right]_{4K} = m_1^4 m_2^4 \sigma_2^4 \Big[\, 1 \;\; \sigma_4^2 \;\; + \sigma_2^2 \left(m_1^2 m_2 + m_2 - 2\right) \sigma_4 \;\; + \sigma_2^4 \left(m_1^3 m_2^3 - m_1^2 m_2 + 2 m_1 m_2 - m_2 - 1\right) \Big]$$

Numerical impact (Gaussian cores, $c = 3$). Ratios true/document of the closed forms:

$(d, N, m, m_1 \times m_2)$	$(2,3,2,2\times1)$	$(2,4,2,2\times1)$	$(2,4,4,2\times2)$	$(2,5,2,2\times1)$	$(2,6,2,2\times1)$	$(3,4,2,2\times1)$
R	1.038	1.513	1.229	2.082	2.899	1.420
$\mathrm{Tr}(M)$	1.000	1.452	1.195	1.975	2.721	1.330

The error compounds per site as $(1 + \Theta(1/m))$.

2 Exact spectral formula

On the constraint $m = m_1 m_2$, $\det \mathbf{W}^{\text{op}} \propto (3\sigma_2^2 - \sigma_4)^2$, so for *Gaussian* cores ($\sigma_4 = 3\sigma_2^2$) the matrix $\mathbf{W}^{\text{op}} = W^{4K} + (d-1)W^{U_1}$ is singular and at most three channels survive; for general c all four may.

Theorem 1 (rank-truncated spectral formula)**.** *With the corrected entries, for every $N \geq 3$,*

$$R(E^N) \;=\; \sum_j \gamma_j \, \mu_j^{N-3}, \qquad \gamma_j = \frac{\left(u_{\text{init}}^{\text{op}\,\top} r_j\right)\left(l_j^\top u_{\text{end}}^{\text{op}}\right)}{l_j^\top r_j},$$

where (μ_j, r_j, l_j) are the eigentriples of $\mathbf{W}^{\text{op}}$, and identically for $\mathrm{Tr}(M)$ with $\mathbf{W}^{\text{Tr}} = d\,W^{4K} + d(d-1)W^A$ and its boundary aggregates. All quantities are explicit rational functions of $(d, m_1, m_2, \sigma_2, \sigma_4)$.

3 Asymptotics

Throughout, $\sigma_2 = 1/m$, $\sigma_4 = c/m^2$. The two extreme splits $(m_1, m_2) \in \{(m,1), (1,m)\}$ give *identical* series below; the balanced split differs only at the orders indicated in Remark 2.

Theorem 2 (Perron data of $\mathbf{W}^{\mathrm{op}}$)**.**

$$\mu_1 = \lambda_*(m) = d + \frac{c-1}{m} + \frac{d\,(cd-c-d+9)}{(d-1)\,m^2} + O(m^{-3}), \qquad \mu_{2,3} = 1 + O(m^{-1}),$$

$$\gamma_1(m) = \frac{2d^4}{(d-1)\,m}\left[1 + \frac{3c(d-1)+d^2-3d+6}{d(d-1)} \cdot \frac{1}{m} + O(m^{-2})\right], \qquad \gamma_2 + \gamma_3 = \Theta(1).$$

For $\mathbf{W}^{\mathrm{Tr}}$*:* $\lambda_*^{\mathrm{Tr}} = d\,\lambda_* + O(m^{-3})$ *and* $\gamma_1^{\mathrm{Tr}} = d^6\,(1 + O(m^{-1}))$*.*

Writing $\beta(m) := \lambda_*/d = 1 + \frac{c-1}{dm} + O(m^{-2})$ for the per-site base,

Corollary 1 (two-regime form)**.**

$$R(E^N) \;=\; \Theta(1) \;+\; \frac{2d}{d-1} \cdot \frac{d^N}{m}\,\beta(m)^{N-3}\big(1 + O(m^{-1})\big), \qquad \mathrm{Tr}(M) \;=\; d^{2N}\beta(m)^{N-3}\big(1 + O(m^{-1})\big),$$

so the universal rank floor $R \geq d^N/m$ *is saturated up to the constant* $\frac{2d}{d-1} \leq 4$ *and the inflation* β^N*; the intrinsic dimension is* $\mathrm{Tr}(M)/R = \Theta\big(\frac{d-1}{2d}\,m\,d^N\big)$ *in the growth regime* $d^N \gtrsim m$*.*

The relative error of the one-channel asymptote decays like $(\mu_2/\lambda_*)^{N-3} \approx d^{-(N-3)}$ (Fig. 1).

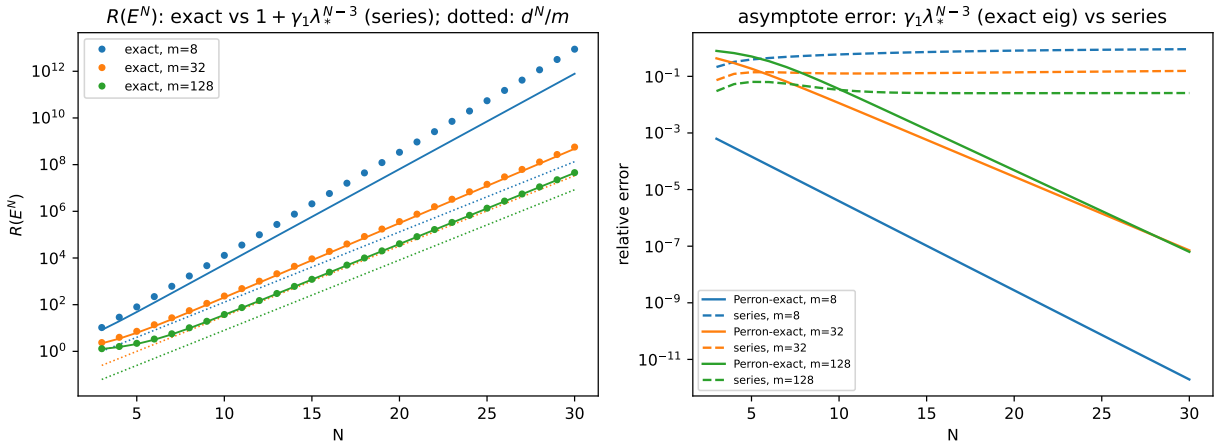


Figure 1: Left: exact $R(E^N)$ (markers) vs the series asymptote $1 + \gamma_1\lambda_*^{N-3}$ (solid) and the floor d^N/m (dotted), $d = 2$, $c = 3$, split $(m, 1)$. Right: relative error of the Perron asymptote with exact eigendata (solid) and with the $O(m^{-2})$ series (dashed).

4 Variance of the k-cycle estimator at $A = I$

With $\mathbb{E}[\hat{E}_k^2]/\,\mathrm{Tr}(A^k)^2 = \mathrm{Tr}(M^k)/d^{2N} \in [\,R^k,\; R^{k-1}\,\mathrm{Tr}(M)\,]/d^{2N}$ and Corollary 1, in the growth regime $d^N \gtrsim m$:

Corollary 2 (explicit variance scaling)**.**

$$\frac{\mathbb{E}[\hat{E}_k^2]}{\mathrm{Tr}(A^k)^2} \;=\; \Theta_d\Big(\,d^{\,N(k-2)}\;m^{-k}\;\beta(m)^{\,Nk}\,\Big), \qquad \beta(m) = 1 + \frac{c-1}{dm} + O(m^{-2}),$$

with explicit bracket: lower constant $\left(\frac{2d}{d-1}\right)^k$*, upper/lower ratio* $\leq \mathrm{Tr}(M)/R = \Theta(m\,d^N)$ *(same exponential base). Consequently a relative accuracy* ε *with* $O(1)$ *samples requires* both

$$(i) \quad m \gtrsim d^{N(1-2/k)}\,\varepsilon^{-2/k} \qquad \text{and} \qquad (ii) \quad m \gtrsim \frac{(c-1)\,Nk}{d},$$

(i) being the rank floor (tight: it matches the universal lower bound), (ii) the inflation control. For $k \leq 2$*, (i) is* $O(\varepsilon^{-2/k})$ *and* $m = \Theta(N)$ *already gives* $\beta^{Nk} \leq e^{(c-1)k/d}$ *— e.g.* $d = 2$*,* $c = 3$*,* $k = 2$*,* $m = N$*:* $\beta^{2N} \to e^2$*, recovering the empirically observed* $O(e^2)$ *variance plateau. For* $k = 3$ *the floor* $d^{N/3}$ *is unavoidable.*

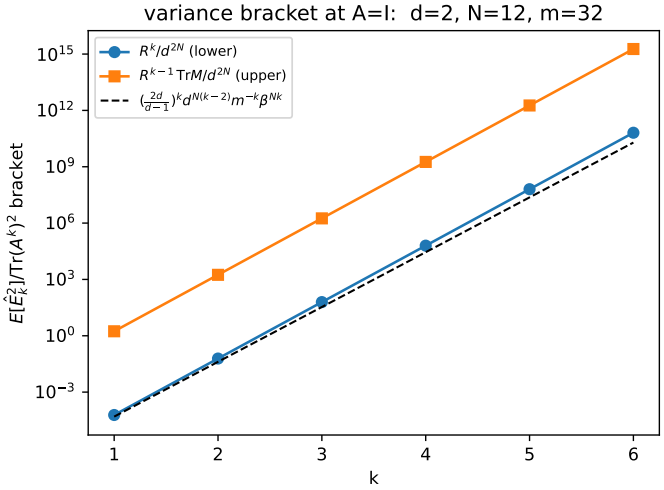


Figure 2: Exact bracket $[R^k, R^{k-1}\,\mathrm{Tr}\,M]/d^{2N}$ vs the formula of Corollary 2 (d=2, N=12, m=32).

5 Design consequences

Remark 1 (fourth-moment design)**.** *The leading inflation is proportional to* $c-1 = \sigma_4/\sigma_2^2 - 1$*. Cores with* $\sigma_4 = \sigma_2^2$ *(e.g. Rademacher entries* $\pm\sigma$*) give*

$$\lambda_* = d + \frac{8d}{(d-1)\,m^2} + O(m^{-3}),$$

i.e. the per-site penalty drops from $\Theta(1/m)$ *to* $\Theta(1/m^2)$ *and requirement (ii) weakens to* $m \gtrsim \sqrt{8Nk/(d-1)}$ *(Fig. 3).*

Remark 2 (gadget split)**.** λ_* *and* λ_*^{Tr} *are split-independent through* $O(m^{-2})$*. Numerically (at* $d = 2$*,* $c = 3$*) the balanced split improves subleading terms only: second-order coefficient* ≈ 5.7 *vs* 20*, and boundary constant* $\kappa_0 \approx 2d^3$ *vs* $2d^4/(d-1)$*.*

6 Verification

`r_en_transcription.py` (document transcription + dense MC of the sketch), `wick_exact.py` (exact 4-replica Wick referee), `true_transfer.py` (symbolic partition-sum engine; matches Wick to 0 and reproduces the chains to 10 digits), `errata.py` (the 33-entry census), `asymptotics3.py`, `asymptotics4.py` (Perron and charpoly-ansatz series, validated numerically to $O(m^{-3})$ residuals), `figures.py`. All entries in Section 1 are emitted directly from the validated engine.

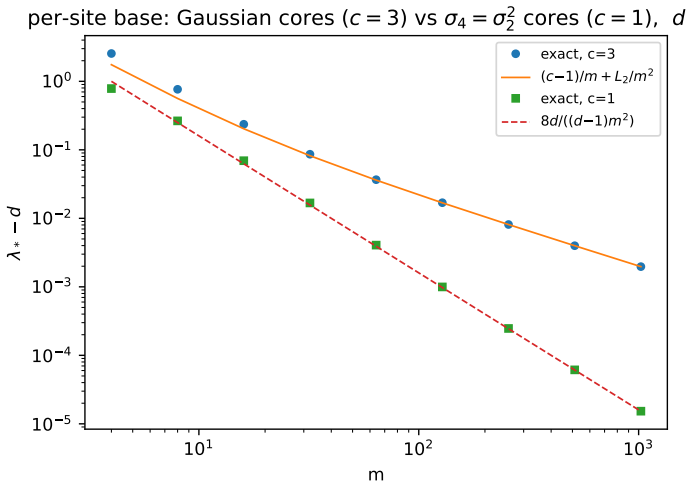


Figure 3: $\lambda_* - d$ vs m: Gaussian (c=3) vs $\sigma_4 = \sigma_2^2$ (c=1) cores, with the series overlays.